

A CONTINUOUS MARKOV-CHAIN PERSPECTIVE ON JOINT SEISMIC RELIABILITY AND RESILIENCE ASSESSMENT

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ABSTRACT

In the current paradigm of the PEER-PBEE (Pacific Earthquake Engineering Research Center Performance-Based Earthquake Engineering) framework [1], recovery processes after seismic events are mostly handled as a post-event post-processing step, rather than being organically integrated into the Poissonian treatment of structural performance. While the paradigm is gradually shifting toward a more comprehensive approach covering downtime, recovery, and resilience, the main limitation in extending the PEER-PBEE framework stems from the intrinsic assumptions of memorylessness and renewability of Poissonian processes [2].

To address these limitations, we propose a generalization of the classical PEER-PBEE framework centered around continuous-time Markov chain (CTMCs, [3]) as the core tool to track the time evolution of the performance state of structures under seismic loading. This generalization relaxes the Poisson assumption and enables a much richer framework that preserves the simplicity and the decoupling properties of the original formulation. Pairing CTMCs with recent advances in state-dependent fragility analysis [4] offers an elegant yet powerful platform to consistently handle both damage and recovery processes. In addition, this novel perspective provides a comprehensive array of analytical and numerical tools that can yield accurate, efficient, and insightful estimates of critical performance indices, such as reliability and resilience indices.

We demonstrate the effectiveness of the proposed methodology through an illustrative synthetic case study based on realistic industrial structures.

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